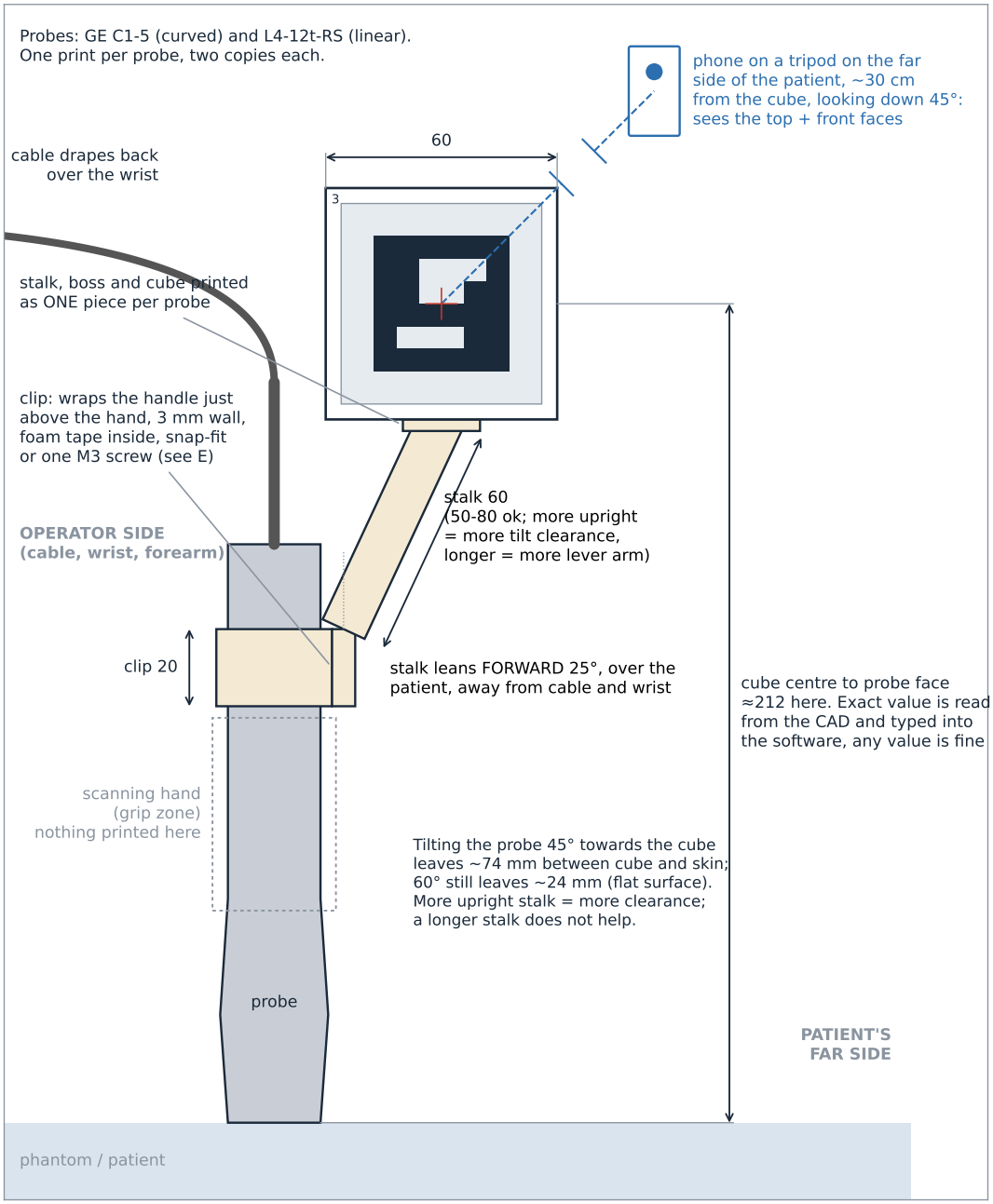
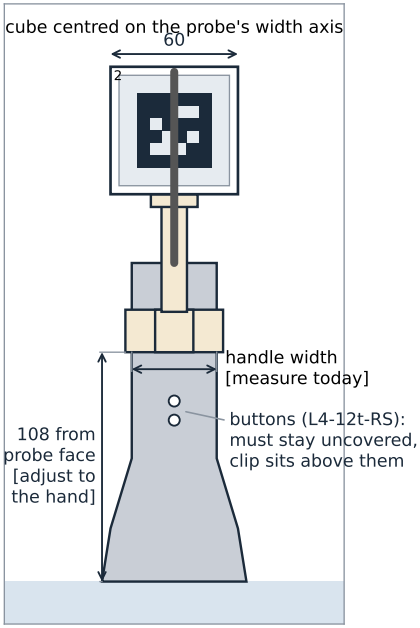
Marker cube mount for the ultrasound probe: print sketch

Ben Zheng, MS2 · for Emily Simpson · 2026-11-09 · all dimensions in mm · page 1 of 2: assembly and cube

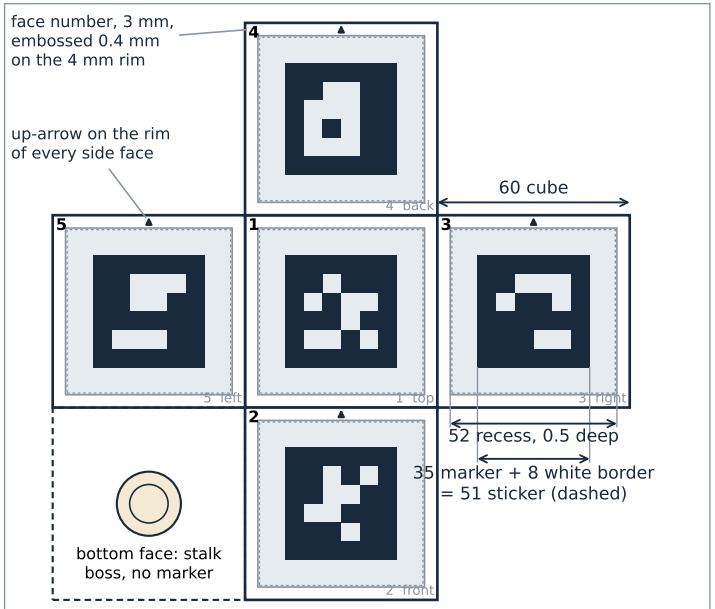
A Side view, probe on the phantom (mm, 1:1 within the panel)



B Front view (button side), stalk towards you



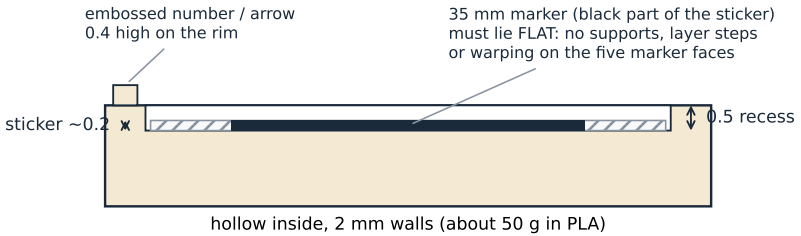
C Cube unfolded: five marker faces + bottom (stalk)



Stickers (markers) are printed on paper by us and glued into the recesses after printing. All five recesses identical; which marker goes on which face is recorded by the face number.

Marker cube mount for the ultrasound probe: print sketch

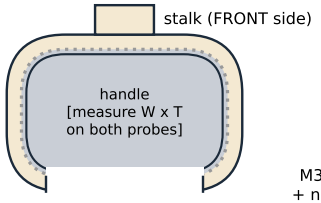
D Section through one face (vertical scale x5)



The recess stops the sticker edge peeling and self-centres it. Faces must be flat and matte white (markers are found by black/white contrast; gloss and dark filament break detection).

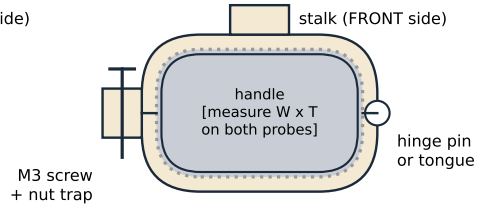
E Clip cross-section at the handle: two ways to make it (pick one)

E1 snap-fit C-clip (simplest, no hardware)
PLA may crack when flexed: PETG helps



opening at the BACK (cable side),
handle width minus ~8 mm; flex to snap on

E2 two-piece clamp, one M3 screw + nut
(firmer, reusable; resized per probe)



Both: 3 mm wall, 20 mm tall, 1 mm foam tape inside (dotted) so the probe is not scratched; cable and buttons untouched; must not rotate or slide once on (any slip moves every measurement). Split line = the two halves.

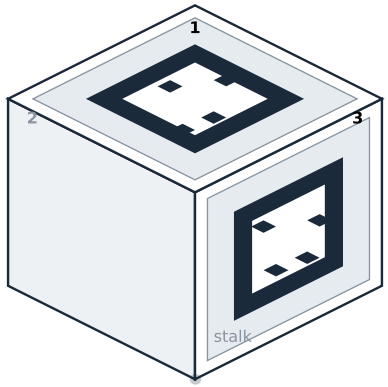
F What to print (summary) and what to measure today

Cube	60 mm edge (55-60 ok), hollow, 2 mm walls, matte WHITE
Marker faces	5 faces (all but bottom): recess 52 x 52 x 0.5 mm, flat, no supports on them
Labels	face numbers 1-5 (3 mm) + up-arrow on side faces, embossed 0.4 mm on the rim
Stalk	60 mm, 12 mm round, leans FORWARD ~25° over the patient (away from cable and wrist)
Clip	above the hand, 20 mm tall, 3 mm wall, M3 clamp (E2) or snap open at the back (E1)
Keep free	the scanning grip, both buttons (L4-12t-RS), the cable exit
Pieces	clip + stalk + cube as ONE part per probe (C1-5, L4-12t-RS); 2 copies each = 4 prints
Material	PLA fine (PETG if snap-fit); white, matte; no glossy or dark filament
Camera	phone ~30 cm away on the far side of the patient, 45° down: top + front face visible
Tolerances	nothing tight: ±0.5 mm is fine everywhere except the clip's grip on the handle

Measure today (bring calipers and a phone):

- handle width and thickness of BOTH probes at 100-130 mm from the probe face (clip zone)
- position of the two buttons on the L4-12t-RS, and the cable-exit diameter
- a photo of each probe with a ruler, side and front, so the handle can be modelled
- Emily's printer, nozzle, filament colours, and the largest part her bed takes (≈ 60 x 90 x 150 mm needed)

G What the phone sees (cube in the camera's eye, 45° down)



Two faces at right angles in every frame is what gives the tracker a stable 3-D pose.
So the cube sits on a stalk in front of and above the hand: fingers never cover a face.
A third face (2) caught at a glancing angle helps too.